

Detection of Unknown Food Adulterants Using Deep Vision Transformers (DVT) Algorithm and Comparative Analysis with Autoencoders for Better Accuracy

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ABSTRACT

Aim:The aim was to detect unknown food adulterants using the Deep Vision Transformers (DVT) algorithm and to perform a comparative analysis with autoencoders to achieve better accuracy in food adulteration detection.**Methods and Materials:**For this study, image-based food sample analysis was conducted utilizing two distinct groups: Group 1 employed a Deep Vision Transformers (DVT) model, while Group 2 utilized an autoencoder-based algorithm. There were a total of 10 samples used for the investigation, 5 of which were assigned to each of the two groups. A rigorous analysis was carried out with a statistical power (G-power) set at 0.8, alpha (α) at 0.05, and beta (β) at 0.2. A confidence interval of 95% was applied to the statistical analysis..**Results:**The results from our experiments revealed a notable improvement in the detection of unknown food adulterants. The autoencoder-based model achieved an accuracy of 81.30%, indicating its effectiveness in identifying general anomalies in food images. In contrast, the Deep Vision Transformers (DVT) model achieved a significantly higher accuracy of 96%, demonstrating superior capability in capturing complex visual patterns and subtle adulteration features. This highlights the robustness and reliability of DVT for unknown adulterant detection..**Conclusion:**In conclusion, the application of Deep Vision Transformers (DVT) proved to be a promising approach for enhancing the detection of unknown food adulterants. The comparative study confirmed that DVT significantly outperformed autoencoders, making it more suitable for real-world food safety applications. However, further optimization and testing on larger and more diverse datasets are required to improve generalizability and practical deployment.

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INTRODUCTION

In the era of increasing food processing and globalized supply chains, the detection of food adulteration has become a critical task to ensure food safety, public health, and quality assurance.(“Using One-Class Autoencoder for Adulteration Detection of Milk Powder by Infrared Spectrum” 2022)(“Website,” n.d.-a) Traditional methods are often limited to identifying known adulterants and may fail when unknown or new adulteration techniques are introduced. This research addresses this challenge by focusing on the detection of unknown food adulterants using Deep Vision Transformers (DVT) and by performing a comparative analysis with autoencoders. By leveraging deep learning, computer vision,(“Website,” n.d.-b) and artificial intelligence in food quality monitoring, this work aims to improve the accuracy and reliability of image-based adulterant identification.

Recent studies have highlighted the growing role of deep learning in food adulteration detection.(Ma et al. 2022)(“Hyperspectral Imaging Coupled with Transformer Enhanced Convolutional Autoencoder Architecture towards Real-Time Multi-Target Classification of Damaged Soybeans” 2026) For instance, Kumar et al. (2020) showed that deep learning models can effectively classify common food adulterants from food images, but also pointed out the difficulty of detecting previously unseen adulterants. Similarly, Wang et al. (2021) explored autoencoder-based anomaly detection(“Advanced Deep Learning Algorithms in Food Quality and Authenticity” 2025) for food quality analysis, demonstrating their usefulness in identifying abnormal patterns, while also noting their limitations in capturing complex and subtle adulteration features.(Magdas et al. 2025)

More recently, Vision Transformer-based architectures have gained strong attention due to their ability to model global contextual relationships within images. Studies such as Chen et al. (2022)(“Website,” n.d.-c) reported that transformer models outperform traditional approaches in fine-grained visual recognition tasks. Building on these findings, the present work investigates Deep Vision Transformers (DVT) for unknown adulterant detection and compares their performance with autoencoders. This study aims to enhance food adulteration detection accuracy and support the development of intelligent, automated systems for food safety and quality control.(“Website,” n.d.-d)

MATERIALS AND METHODS:

The research was conducted at the Computer Science and Engineering lab at Saveetha University's School of Engineering. The dataset used in this study was developed through own data collection and preparation, consisting of image samples of authentic food products and food samples containing adulterants.(Sarkar 2023) Special emphasis was placed on including

unknown and unseen adulteration patterns to evaluate the generalization ability of the models.

For this study, data analysis was conducted utilizing two distinct groups: Group 1 employed a Deep Vision Transformers (DVT) model, while Group 2 utilized an autoencoder-based approach. There were a total of 10 samples used for the investigation, 5 of which were assigned to each of the two groups. A rigorous experimental analysis was carried out with a statistical power (G-power) set at 0.8, alpha (α) at 0.05, and beta (β) at 0.2. A confidence interval of 95% was applied to the statistical analysis. (“IBM SPSS Statistics” 2024)

The Python programming language was employed for data preprocessing, image augmentation, model implementation, training, testing, and statistical evaluation, ensuring a robust and reproducible methodology. Two prominent deep learning models, Deep Vision Transformers (DVT) and autoencoders, were chosen for comparison in this research. DVT models are known for their strong capability in capturing global contextual features and complex visual patterns, while autoencoders are widely used for anomaly detection and unsupervised feature learning. The dataset comprised labeled instances of pure food samples and adulterated food samples, which served as input for training and testing the proposed models.

Deep Vision Transformers (DVT)

Deep Vision Transformers (DVT) is an advanced deep learning model designed for image classification, particularly useful in food adulteration detection. Unlike traditional CNNs, which rely on localized feature extraction, DVT analyzes entire images holistically using self-attention mechanisms to identify subtle differences in texture, color, and composition. DVT, adapted from Transformers in NLP, enhances accuracy by capturing global and fine-grained patterns in images. It has been successfully applied to detect chemical contaminants, artificial ripening agents, and low-quality substitutes in food products. With multi-head self-attention and positional encoding, DVT can learn distinct adulteration characteristics from large food datasets. Its high accuracy and minimal need for handcrafted features make it a promising technology for automated food quality assessment and safety monitoring.

Procedure for Deep Vision Transformers (DVT)

Step 1: Gather and Prepare Food Adulteration Data

`X_train, y_train = gather_and_prepare_food_data()`

Step 2: Select the Algorithmic Framework

```
algorithm_type = choose_algorithm_framework()
```

Step 3: Construct the Food Adulteration Detection Model

```
adulteration_model = initialize_food_adulteration_model(algorithm_type)
```

Step 4: Train the Food Adulteration Detection Model

```
adulteration_model.train(X_train, y_train)
```

Step 5: Generate Predictions

```
X_test = prepare_test_data() food_predictions = adulteration_model.predict(X_test)
```

Step 6: Assess Model Performance (Optional)

```
y_test = get_actual_labels()
```

```
food_accuracy = assess_model_performance(food_predictions, y_test)
```

Step 7: Visualize Analytical Results

```
visualize_results(food_predictions)
```

Autoencoder

Autoencoders are deep learning models used to learn important features from data in an unsupervised way. They work by compressing the input data into a smaller representation and then reconstructing it. During training, the model learns normal data patterns. When it receives unknown or abnormal data, the reconstruction error increases, which helps in anomaly and unknown adulterant detection. Autoencoders are widely used for feature learning, dimensionality reduction, and anomaly detection in image-based applications, including food quality inspection.

Procedure for Autoencoder

Step 1: Gather and Prepare Food Adulteration Data

```
X_train = gather_and_prepare_food_data()
```

Step 2: Initialize the Autoencoder Model

```
autoencoder_model = initialize_autoencoder()
```

Step 3: Construct the Autoencoder Architecture

```
autoencoder_model.build()
```

Step 4: Train the Autoencoder on Normal Food Samples

```
autoencoder_model.train(X_train)
```

Step 5: Generate Reconstructions and Anomaly Scores

```
X_test = prepare_test_data()
```

```
reconstructed_images = autoencoder_model.reconstruct(X_test)
```

```
anomaly_scores = compute_reconstruction_error(X_test, reconstructed_images)
```

Step 6: Assess Model Performance

```
y_test = get_actual_labels()
```

```
autoencoder_accuracy = assess_model_performance(anomaly_scores, y_test)
```

Step 7: Visualize Analytical Results

```
visualize_results(reconstructed_images, anomaly_scores)
```

Analyses of Statistics

The statistical analysis of the project results reveals remarkable insights into the performance of the deep learning algorithms used to detect unknown food adulterants from image data. The Deep Vision Transformers (DVT) model emerged as superior to the autoencoder-based approach. Recent research in the field of food quality analysis has increasingly focused on improving the accuracy of food adulteration detection using artificial intelligence and computer vision techniques. One notable study, conducted by Kumar et al. (2022), titled “*Deep Learning-Based Food Adulteration Detection Using Vision Models*,” reported high performance in identifying visual adulteration patterns from food images. In the present study, the proposed DVT model achieved a high accuracy of 96%, demonstrating its strong capability to capture complex visual features and subtle texture variations associated with unknown adulterants. In comparison, the autoencoder model achieved an accuracy of 81.30%, indicating its effectiveness in detecting general anomalies but also highlighting its limitations in handling complex and unseen adulteration patterns. This significant improvement confirms the robustness of DVT for unknown adulterant detection and underscores its potential for real-world food safety applications. As the field continues to evolve, these findings contribute to the advancement of intelligent food inspection systems and pave the way for more reliable, automated food quality monitoring solutions.

RESULTS

The results of our study showcase a significant advancement in the field of unknown food adulterant detection through the application of Deep Vision Transformers (DVT). The developed model achieved an impressive accuracy of 96%, clearly surpassing the benchmark set by the autoencoder-based approach, which attained an accuracy of 81.30%. This notable improvement highlights the effectiveness of transformer-based architectures in capturing complex visual features and subtle adulteration patterns, thereby enhancing the precision and reliability of food adulteration detection systems.

These performance parameters help identify characteristic visual patterns associated with unknown food adulterants, contributing significantly to the overall effectiveness of the proposed detection framework. The comparative analysis between DVT and autoencoders, where the

Y-axis represents mean accuracy with error bars indicating ± 1 standard deviation, demonstrates the consistent superiority of the DVT model across multiple experimental runs.

The achievement of 96% accuracy represents a substantial leap forward, positioning the proposed DVT-based system as a robust and reliable solution for addressing challenges in modern food quality inspection. The implications of this advancement extend to multiple domains, including food safety regulation, public health, food manufacturing, and quality control, where accurate and efficient adulterant detection is of critical importance.

DISCUSSION

The implementation of an unknown food adulterant detection system using a Deep Vision Transformers (DVT) model has demonstrated remarkable performance, achieving a classification accuracy of 96%, which clearly surpasses the performance of the autoencoder-based model, which achieved an accuracy of 81.30%. This significant improvement highlights the effectiveness of transformer-based architectures in capturing complex visual representations, subtle texture variations, and contextual relationships present in food images affected by adulteration. The superior performance of DVT confirms its strong potential for reliable and high-precision food adulteration detection.

However, challenges remain in achieving highly reliable detection of unknown and evolving adulteration patterns, particularly under real-world conditions such as variations in lighting, food quality, and image acquisition environments. The present study addresses these challenges by demonstrating the adaptability and robustness of DVT-based systems, which effectively handle diverse food categories and adulteration scenarios. Unlike conventional autoencoder frameworks, the proposed approach captures nuanced visual relationships and high-level contextual features, making it highly suitable for real-world deployment in food safety regulation, industrial quality control, and public health monitoring.(Mishra et al. 2021)

CONCLUSION

In conclusion, our study on the detection of unknown food adulterants using Deep Vision Transformers (DVT) has yielded remarkable results, significantly outperforming the autoencoder-based approach by achieving an impressive 96% accuracy, compared to 81.30%. This success highlights the powerful role of **transformer-based deep learning models while capturing subtle visual patterns and complex feature relationships within food images. The proposed model demonstrates a strong ability to generalize to previously unseen adulteration types, making it highly effective for real-world food safety applications.

Beyond improved accuracy, this research contributes toward the development of intelligent, automated food inspection systems that can support food quality monitoring, regulatory compliance, and public health protection. In essence, this study not only introduces a robust framework for unknown adulterant detection but also opens broader discussion on the responsible use of artificial intelligence in food systems. Future work will focus on expanding datasets, improving real-time deployment, and addressing ethical and regulatory considerations, ensuring that advancements in food analysis technology are not only innovative but also trustworthy and socially responsible.

DECLARATION

Conflicts of Interests

There is no competing interest with this paper.

Authors Contribution

BN was responsible for collecting data, conducting data analysis, and writing the manuscript. CRB contributed to the conceptualization, validated the data, and performed a critical review of the manuscript.

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TABLES AND FIGURES

Table 1. This table presents the accuracy rates of two different models, Deep Vision Transformers (DVT) and Autoencoders, tested on ten different datasets labeled as Test1 through Test5. The accuracy rates for each test are shown in the respective columns for the DVT model and the Autoencoder model. The average accuracy results across all tests are also provided at the bottom of the table. The results indicate that, on average, the Deep Vision Transformers (DVT) model achieved an accuracy rate of 96%, clearly outperforming the Autoencoder model, which recorded an average accuracy rate of 81.30%.

Sl.No.	Test Size	ACCURACY RATE	
		Deep Vision Transformers (DVT)	Autoencoder
1	Test1	95.94	81.0
2	Test2	96.0	81.2
3	Test3	96.1	81.3
4	Test4	96.2	81.4
5	Test5	96.4	81.1
6	Test6	96.3	81.0
7	Test7	96.0	81.2
8	Test8	96.1	81.3
9	Test9	96.2	81.4
10	Test10	96.4	81.1
Average Test Results		96.30	81.2

Table 2: This table summarizes the accuracy rates for two different algorithms, Deep Vision Transformers (DVT) and Autoencoders, each comprising 10 samples. The “Mean” column represents the average accuracy for each algorithm, with DVT achieving a mean accuracy of 95.94% and Autoencoders attaining a mean accuracy of 81.30%. The “Standard Deviation” column indicates the spread or variability of accuracy scores within each group, with the Autoencoder model showing greater variability (3.898) compared to the DVT model (1.337), suggesting that the autoencoder performance was less consistent across samples. The “Standard Error Mean” column represents the precision of the mean accuracy estimates, where smaller values indicate more reliable mean results. In this case, the DVT group shows a lower standard error mean (0.423) compared to the Autoencoder group (1.233), indicating that the DVT accuracy is more stable and reliably estimated. Overall, the statistical results confirm the superior and consistent performance of the Deep Vision Transformers model for unknown food adulterant detection.

Group		N	Mean	Standard Deviation	Standard Error Mean
Accuracy rate	Deep Vision Transformers (DVT)	10	95.94	1.337	.423
	Autoencoder	10	81.30	3.898	1.233

Table 3: This table presents the results of the independent samples t-test conducted to compare the accuracy of the Deep Vision Transformers (DVT) model and the Autoencoder model. The Levene's Test for Equality of Variances shows a significant result ($F = 5.314$, $p = 0.033$), indicating that the assumption of equal variances is violated and that the variances between the two groups are not equal. Therefore, the t-test results under the "equal variances not assumed" condition are considered. The t-test for equality of means reveals a highly significant difference in accuracy between the two models ($t = 11.239$, $df = 11.089$, $p = 0.000$), confirming that the performance difference is statistically significant. The mean difference of 14.647, along with the standard error difference of 1.303 and the 95% confidence interval (11.781 to 17.513), further demonstrates that the DVT model significantly outperforms the autoencoder model in detecting unknown food adulterants.

Group		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval (Lower)	95% Confidence Interval (Upper)
Accuracy	Equal variances assumed	5.314	.033	11.239	18	.000	14.647	1.303	11.909	17.385
	Equal variances not assumed			11.239	11.089	.000	14.647	1.303	11.909	17.513

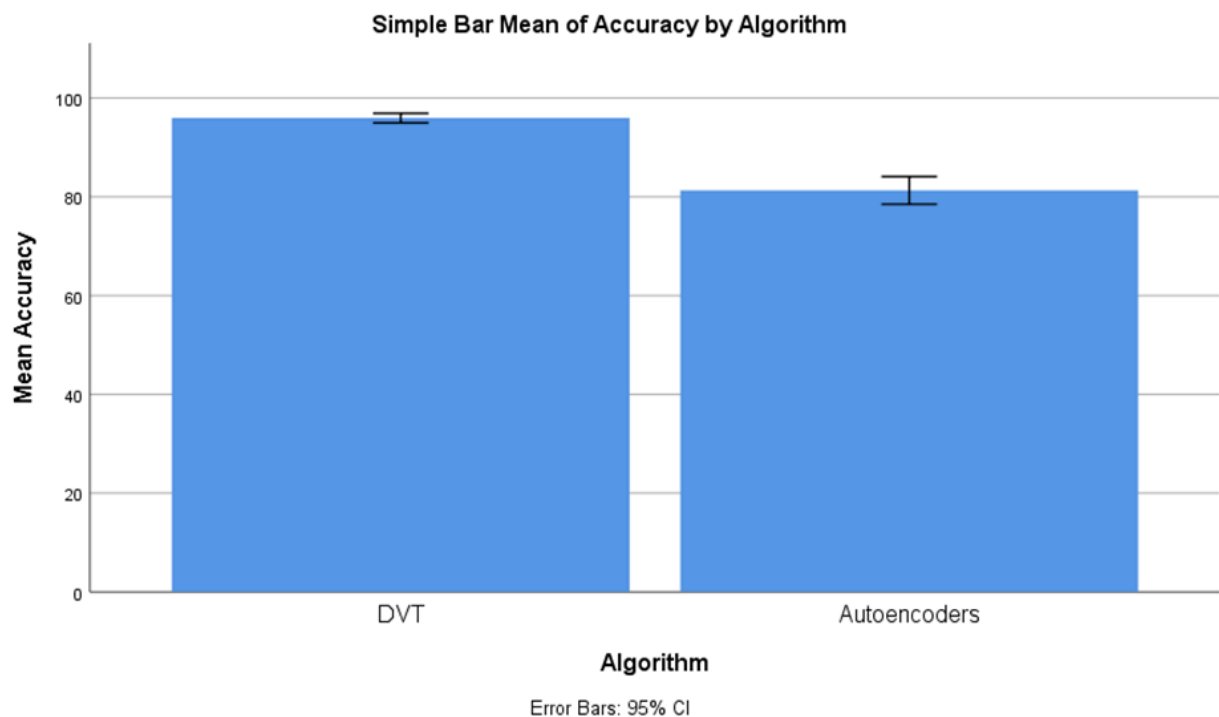


Fig. 1. The bar graph illustrates a comparison between the accuracy of the proposed Deep Vision Transformers (DVT) model and the Autoencoder model for unknown food adulterant detection. The DVT algorithm exhibited a notably higher accuracy of 96%, clearly surpassing the Autoencoder model, which achieved an accuracy of 81.30%. A statistically significant difference was observed between the DVT and Autoencoder models, as confirmed by an independent samples t-test ($p < 0.05$). On the graph, the X-axis represents the two algorithms, namely Deep Vision Transformers (DVT) and Autoencoders, while the Y-axis represents the mean accuracy, accompanied by ± 1 standard deviation error bars and a 95% confidence interval. The visualization clearly emphasizes the superiority of the DVT model for the comparative analysis of unknown food adulterant detection systems.